

Challenge (Habanero)

- Q1.** A construction company is building a rectangular foundation with an area of $x^2 + 5x + 6$ square meters. What are the factors of the area?
- (A) $x^2 + 5x + 6 = (x + 2)(x + 3)$
(B) $x^2 + 5x + 6 = (x - 2)(x - 3)$
(C) $x^2 + 5x + 6 = (x + 1)(x + 4)$
(D) $x^2 + 5x + 6 = (x - 1)(x - 4)$
(E) $x^2 + 5x + 6 = (x + 3)(x - 2)$
- Q2.** A recipe for cookies requires $x^2 - 7x - 18$ cups of flour. What are the factors of the amount of flour?
- (A) $x^2 - 7x - 18 = (x - 9)(x + 2)$
(B) $x^2 - 7x - 18 = (x + 9)(x - 2)$
(C) $x^2 - 7x - 18 = (x - 6)(x - 3)$
(D) $x^2 - 7x - 18 = (x + 6)(x + 3)$
(E) $x^2 - 7x - 18 = (x - 2)(x + 9)$
- Q3.** A architect is designing a rectangular building with a perimeter of $2x^2 + 5x + 3$ meters. What are the factors of the perimeter?
- (A) $2x^2 + 5x + 3 = (2x + 1)(x + 3)$
(B) $2x^2 + 5x + 3 = (2x - 1)(x - 3)$
(C) $2x^2 + 5x + 3 = (2x + 3)(x + 1)$
(D) $2x^2 + 5x + 3 = (2x - 3)(x - 1)$
(E) $2x^2 + 5x + 3 = (x + 1)(2x + 3)$
- Q4.** A bakery is making a cake that requires $x^2 - 4x - 5$ teaspoons of sugar. What are the factors of the amount of sugar?
- (A) $x^2 - 4x - 5 = (x - 5)(x + 1)$
(B) $x^2 - 4x - 5 = (x + 5)(x - 1)$
(C) $x^2 - 4x - 5 = (x - 1)(x + 5)$
(D) $x^2 - 4x - 5 = (x + 1)(x - 5)$
(E) $x^2 - 4x - 5 = (x + 4)(x - 1)$
- Q5.** A construction site has a rectangular pit with an area of $x^2 + 2x - 15$ square meters. What are the factors of the area?
- (A) $x^2 + 2x - 15 = (x + 5)(x - 3)$
(B) $x^2 + 2x - 15 = (x - 5)(x + 3)$
(C) $x^2 + 2x - 15 = (x + 3)(x - 5)$
(D) $x^2 + 2x - 15 = (x - 1)(x + 15)$
(E) $x^2 + 2x - 15 = (x + 1)(x - 15)$
- Q6.** A recipe for bread requires $x^2 + x - 12$ cups of flour. What are the factors of the amount of flour?
- (A) $x^2 + x - 12 = (x + 4)(x - 3)$
(B) $x^2 + x - 12 = (x - 4)(x + 3)$
(C) $x^2 + x - 12 = (x + 3)(x - 4)$
(D) $x^2 + x - 12 = (x + 1)(x - 12)$
(E) $x^2 + x - 12 = (x - 1)(x + 12)$
- Q7.** A building has a rectangular foundation with a perimeter of $2x^2 - 5x - 3$ meters. What are the factors of the perimeter?
- (A) $2x^2 - 5x - 3 = (2x + 1)(x - 3)$
(B) $2x^2 - 5x - 3 = (2x - 1)(x + 3)$
(C) $2x^2 - 5x - 3 = (x + 1)(2x - 3)$
(D) $2x^2 - 5x - 3 = (x - 1)(2x + 3)$
(E) $2x^2 - 5x - 3 = (2x + 3)(x - 1)$
- Q8.** A recipe for cake requires $x^2 - 9x + 20$ cups of sugar. What are the factors of the amount of sugar?
- (A) $x^2 - 9x + 20 = (x - 5)(x - 4)$
(B) $x^2 - 9x + 20 = (x + 5)(x + 4)$
(C) $x^2 - 9x + 20 = (x + 4)(x - 5)$
(D) $x^2 - 9x + 20 = (x - 4)(x + 5)$
(E) $x^2 - 9x + 20 = (x - 1)(x + 20)$

Open Ended Questions (FRQ)

Q9. Solve the equation $x^2 + 4x - 5 = 0$ by factoring and verify the solutions.

Q10. Solve the equation $x^2 - 6x + 8 = 0$ by factoring and verify the solutions.

Chef's Tips

- Check for factorization and solutions
- Verify the work and reasonableness of answers
- Provide feedback on problem-solving strategies